

SDT

Self Development Tracker

From a personal challenge idea to a working full-stack product — approached through a Technical Business Analyst lens.

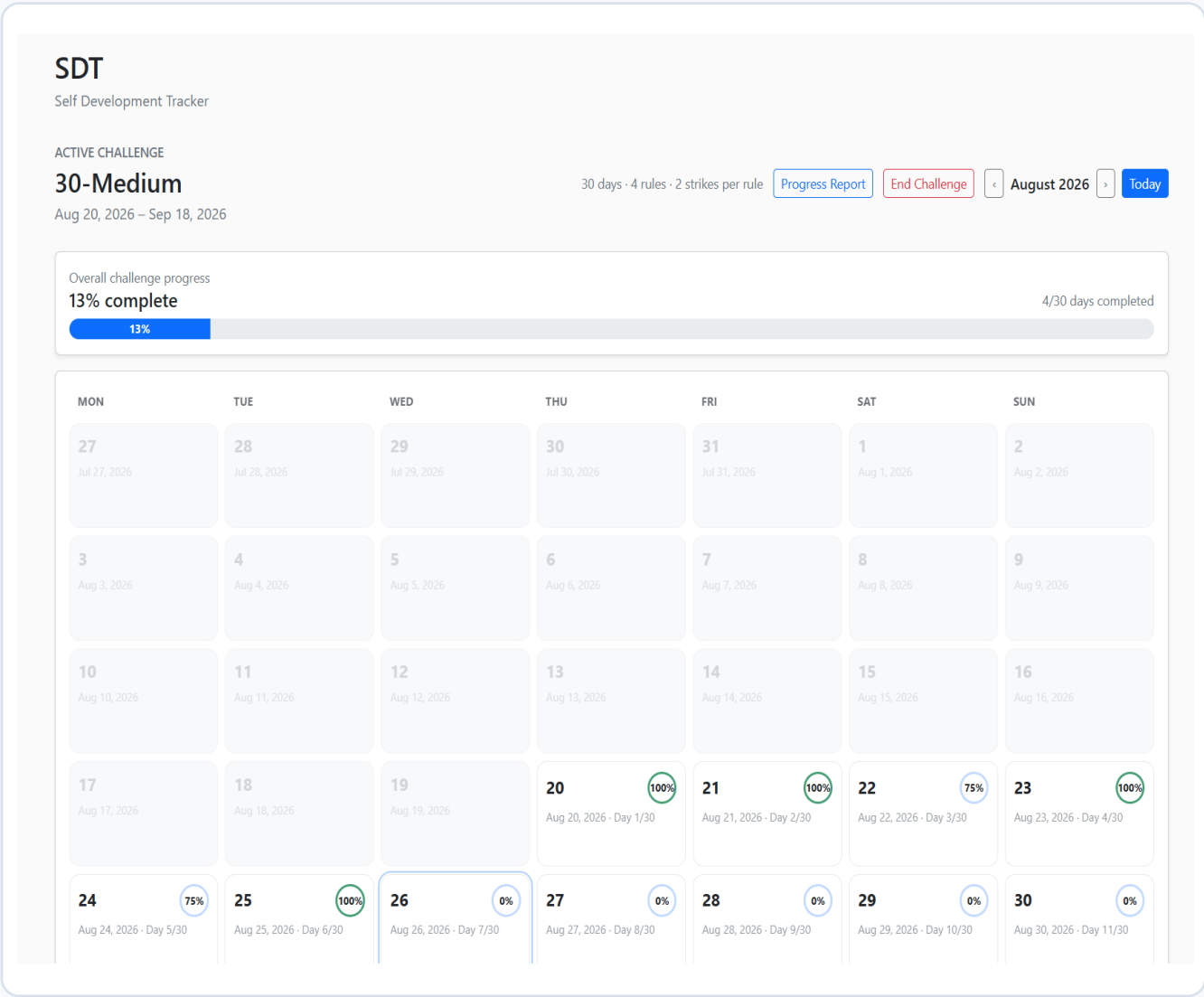
BA CASE STUDY

FULL-STACK

LOCAL V1

Problem → Requirements → Design → Build → Test → Improve

Portfolio case study • SDT V1 • Kyaw Zay Han • 29 August 2026



01

The problem I wanted to solve

Structured challenges are simple to understand, but surprisingly hard to track consistently.

The gap

- Daily rules are easy to forget or record inconsistently.
- A basic habit checklist does not show the full challenge lifecycle.
- Missed tasks, strikes and unrecorded days need different treatment.
- Progress needs to be understandable at both day and rule level.

Design question

How can I turn a fixed-duration self-development challenge into a clear, auditable workflow that is fast enough to use every day?

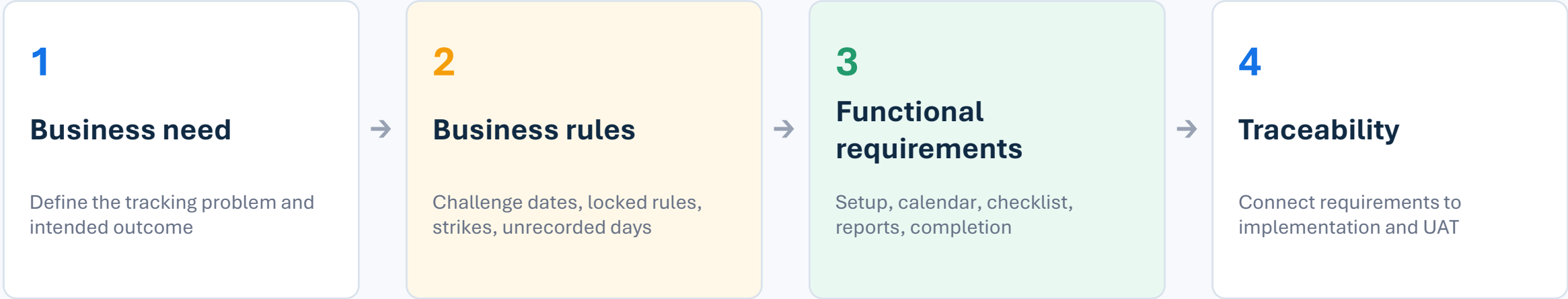
Success looked like

- Configure once before starting
- Record today in seconds
- Keep unrecorded ≠ failed
- See progress and strikes clearly
- Generate usable progress/final reports

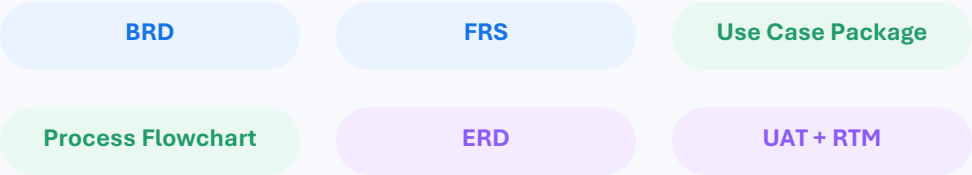
02

Turning the idea into requirements

I treated the project as a small product lifecycle, not just a coding exercise.



BA artifacts produced



This made the project easier to explain, test and change because each feature had a reason for existing.

03

The solution: one challenge lifecycle

SDT keeps setup, daily tracking, monitoring and completion inside one workflow.

SDT

Self Development Tracker

Build Your Challenge

Define the commitment before you begin. Once the challenge starts, its rules are locked.

Challenge Settings

Challenge name

30-Medium

Start date

08/20/2026

Challenge duration

30

days

Strikes allowed per rule

2

Rules & Tasks

These will become your daily checklist.

+ Add Rule

1

Cold Shower

↑

↓

Remove

2

Hit the gym

↑

↓

Remove

3

Read 10 pages

↑

↓

Remove

4

Send 20 quality job applications

↑

↓

Remove

Start Challenge

Configure

Name, duration, start date, strike allowance and ordered rules.

Track

Calendar + Today shortcut opens the daily checklist.

Monitor

Completion, rule-level misses and remaining strikes.

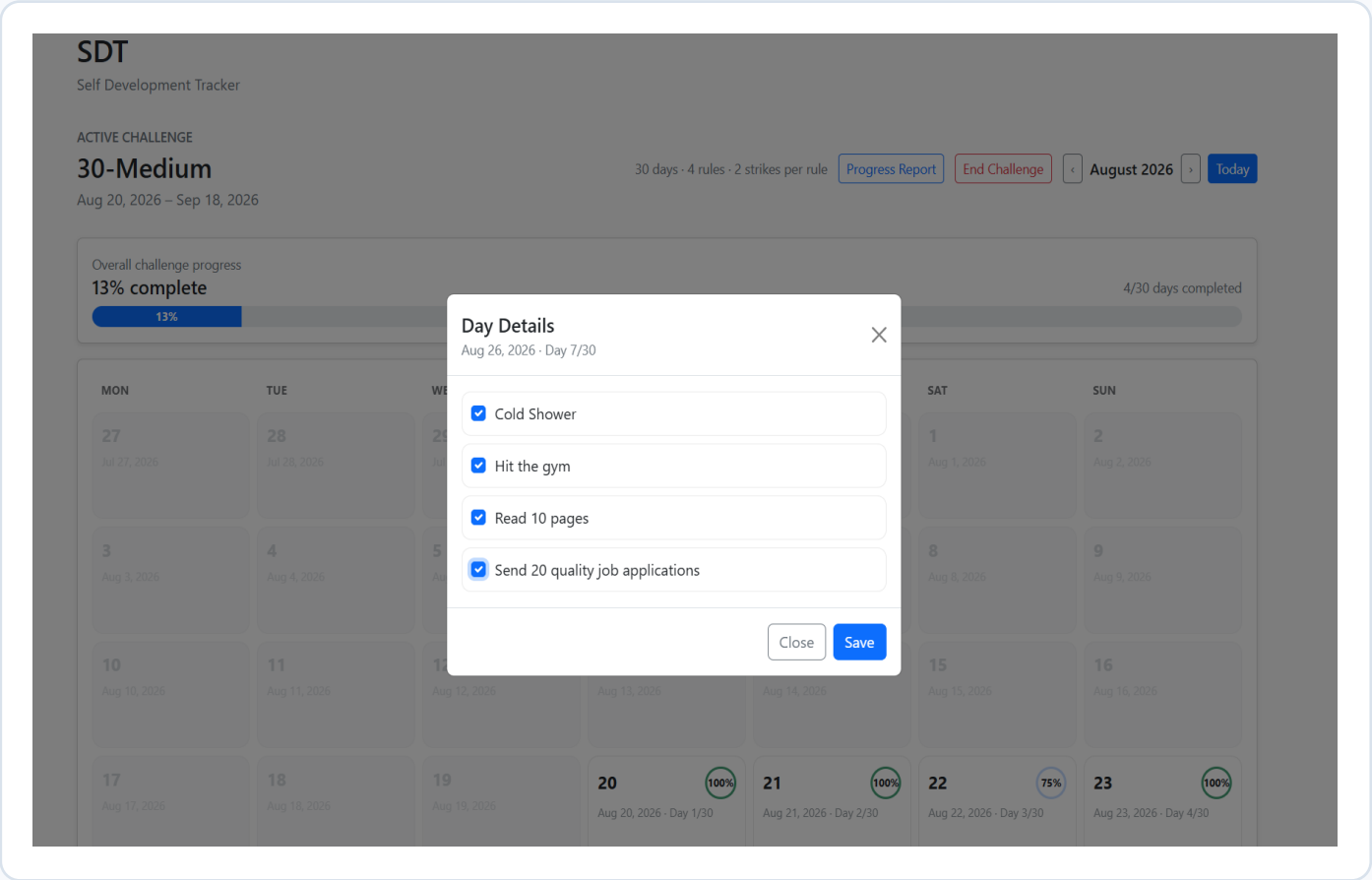
Report

In-progress and final challenge reporting with print output.

04

Daily tracking designed for low friction

The most frequent action should require the fewest clicks.



Key interaction decisions

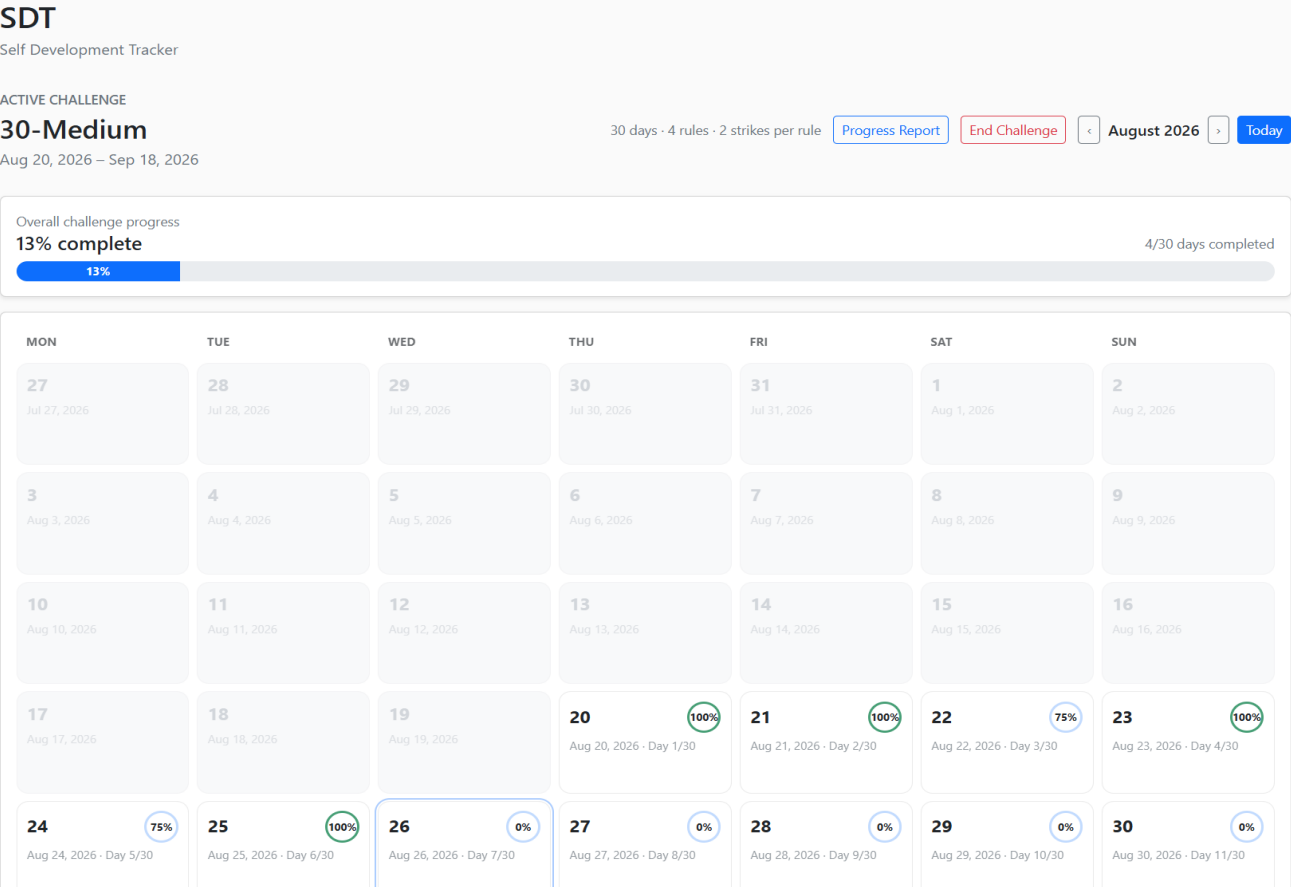
- Today button acts as a shortcut to the current checklist.
- Tick / untick tasks, then Save; the dialog closes automatically.
- Past eligible dates remain editable from the calendar.
- Dates outside the challenge window are disabled.

Business rule An unrecorded day is not automatically treated as a failed day.

05

Monitoring without overcomplicating the dashboard

The dashboard separates challenge progress, daily completion and rule-level strikes.



What the user can see

- Overall perfect-day progress
- Daily percentage by calendar date
- Recorded vs. unrecorded state

Strike logic

Strikes are informational. The tracker shows used / remaining strikes per rule without forcing a reset or penalty workflow.

06

Reporting turns daily records into insight

Progress reporting was built from the same underlying challenge and daily-status data.

SDT

Challenge Progress Report

Back to SDT

Print Report

ACTIVE CHALLENGE

30-Medium

Aug 20, 2026 – Sep 18, 2026 · 30 days · 4 rules

Generated Aug 26, 2026, 2:32 PM

Progress Overview

Current Day

7/30

Recorded Days

6/7

Perfect Days

4

Avg. Recorded Completion

92%

Completed Tasks

22

Missed Tasks

2

Unrecorded Days

1

Strikes Used

2

Challenge Completion

13% of the full challenge completed as perfect days (4/30). 23 planned day(s) remain.

Performance Highlights

STRONGEST RULE

Read 10 pages

100% completion · 6/6 recorded days completed

WEAKEST RULE

Cold Shower

83% completion · 5/6 recorded days completed

Rule Performance

Completed

Missed

Completion

Strikes

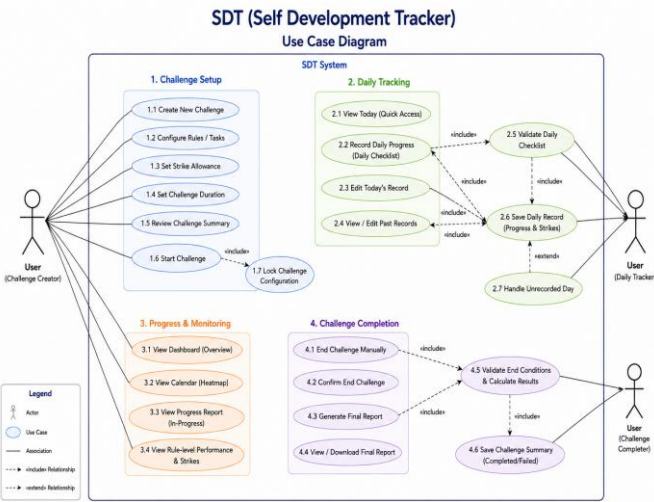
Current day • recorded days • perfect days • average completion • strikes

Performance Highlights				
STRONGEST RULE Read 10 pages 100% completion · 6/6 recorded days completed		WEAKEST RULE Cold Shower 83% completion · 5/6 recorded days completed		
Rule Performance				
Rule / Task	Completed	Missed	Completion	Strikes
Cold Shower	5/6	1	83%	1/2
Hit the gym	5/6	1	83%	1/2
Read 10 pages	6/6	0	100%	0/2
Send 20 quality job applications	6/6	0	100%	0/2
Daily Performance				
Day	Date	Status	Completed	Completion
Day 1	Aug 20, 2026	Perfect	4/4	100%
Day 2	Aug 21, 2026	Perfect	4/4	100%
Day 3	Aug 22, 2026	Recorded	3/4	75%
Day 4	Aug 23, 2026	Perfect	4/4	100%
Day 5	Aug 24, 2026	Recorded	3/4	75%
Day 6	Aug 25, 2026	Perfect	4/4	100%
Day 7	Aug 26, 2026	Unrecorded	—	—

From requirements to system design

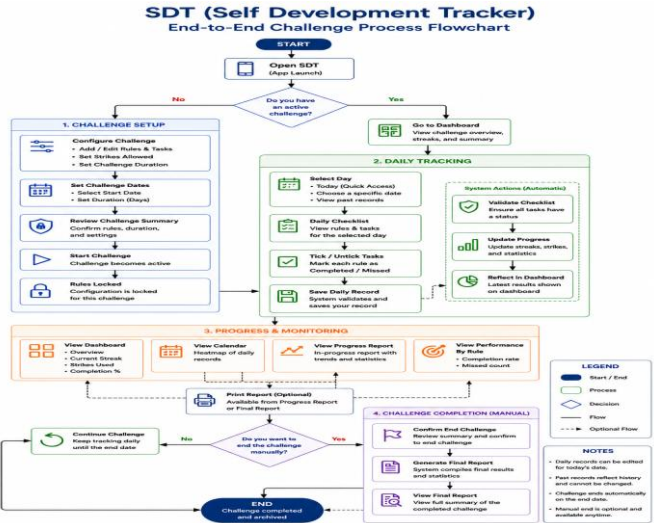
The BA artifacts show how user needs translate into process, behaviour and data.

Use Case Model



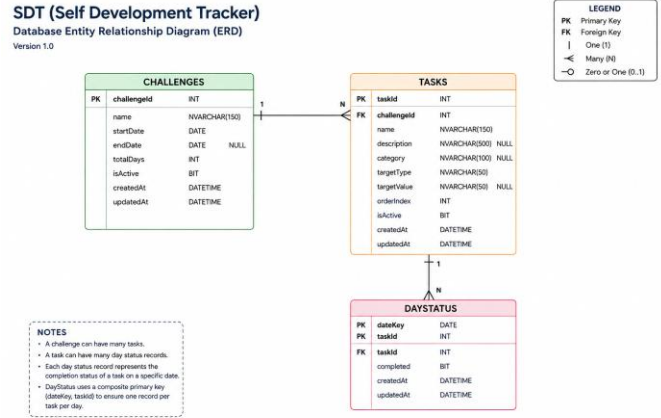
Who interacts with each capability and how behaviours relate.

Process Flow



The end-to-end challenge lifecycle and key decisions.

Database ERD



How challenge, task and daily-status data are structured.

08

Build, test and document

I used the documentation as a working bridge between intended behaviour and the implemented product.

Front end

HTML • CSS • Bootstrap-style UI • Vanilla JS

Back end

Node.js • Express • REST endpoints

Data

Microsoft SQL Server • parameterised queries

Local delivery

localhost:4000 • .env configuration • one-click .bat launcher

Validation approach

- Manual functional testing during development
- Lightweight UAT test pack for final acceptance
- Final traceability links requirements → feature → UAT evidence
- README + first-time local installation guide

V1 status

Working local application with the core challenge lifecycle implemented and documented.

09

Outcome and next iteration

The next step is not more documentation — it is making the product easier for other people to try.

What V1 demonstrates

- Requirements translated into a working product
- Business rules reflected in actual system behaviour
- Process + data modelling alongside development
- Testing and traceability before sign-off
- Documentation for another user to run it locally

Next iteration

Feedback: local setup creates unnecessary friction for testers.

Planned response

Deploy a cloud-accessible version so friends can open and test SDT without installing Node.js or SQL Server locally.

This becomes the next product iteration: feedback → prioritisation → deployment → new user testing.

SDT

A small project, built as a complete product lifecycle.

Problem framing • Requirements • Process modelling • System design • Development • Testing • Iteration

Technical Business Analyst portfolio case study

SDT V1 — Self Development Tracker

Next: cloud deployment and external user feedback

GitHub Link @ <https://github.com/kyawzayhan26/Self-Development-Tracker-SDT-local.git>